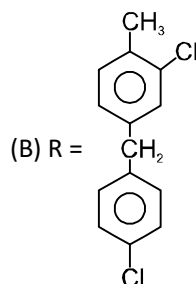
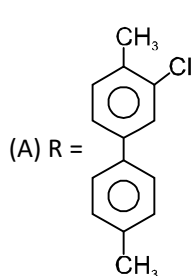
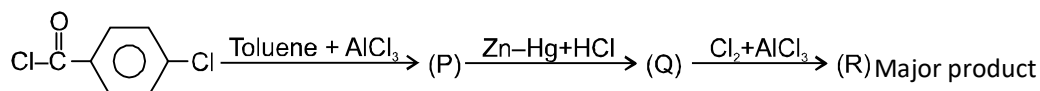
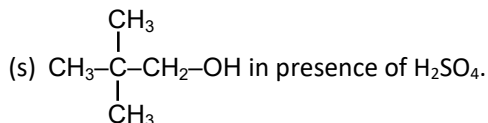
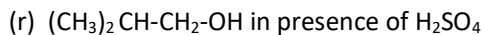
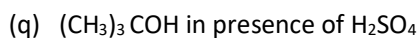
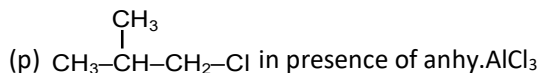


SCQ (Single Correct Type) :

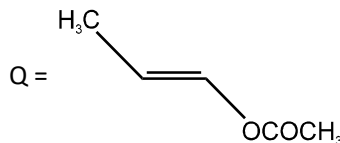
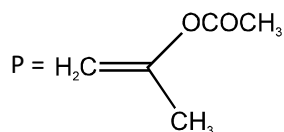
1. Identify incorrect statements for given reaction sequence.



- (C) In the given three reaction steps, two are electrophilic substitution steps.
 (D) Product(P) gives a racemic mixture with (i) MeMgBr (ii) aq. NH_4Cl
2. Which of the following reagents used for alkylation of benzene by electrophilic substitution mechanism will give same product ?

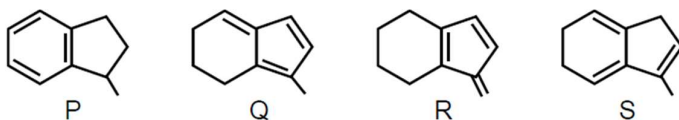


- (A) p, q, r (B) q, r, s (C) p, q, r, s (D) p, q, s
3. The product of acid hydrolysis of P and Q can be distinguished by :



- (a) Benedictt solution (b) 2,4-DNP (c) Fehling's solution (d) Tollen reagent
 (A) (b) & (d) (B) (a), (b) & (c) (C) (a) & (c) (D) (a), (c), (d)

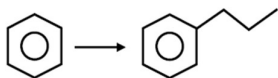
4. Which of the following is incorrect for the following set of compounds?



- (A) P gives phthalic acid on treatment with hot alkaline KMnO_4/H^+ .
 (B) Q is the most acidic compound among given hydrocarbons
 (C) The general formula of compound R is $\text{C}_n\text{H}_{2n-8}$.
 (D) Geometrical isomerism is possible in S

MCQ (Multiple Correct Type) :

5. Below transformation of benzene to propyl benzene is carried out by which of the following reactions?



- (A) + $\xrightarrow{\text{AlCl}_3}$
- (B) + $\text{CH}_3-\text{CH}=\text{CH}_2 \xrightarrow{\text{H}^+}$
- (C) + $\text{CH}_3-\text{CH}_2-\text{COCl} \xrightarrow[\text{(ii) Zn-Hg, Conc. HCl}]{\text{(i) AlCl}_3}$
- (D) + $\text{CH}_3-\text{CH}_2-\text{C}(=\text{O})-\text{Br} \xrightarrow[\text{(ii) N}_2\text{H}_4 \cdot \text{OH}^-]{\text{(i) AlCl}_3}$

6. Select the correct statement(s)

(A) Formic acid is stronger acid than benzoic acid

(B) Compound show Keto-enol tautomerism

(C) Phenol & benzoic acid can be distinguished by NaHCO_3 .

(D) Order of basicity in aqueous medium $\text{MeNH}_2 < \text{Me}_2\text{NH} < \text{Me}_3\text{N}$

7. Which of the following reactions take place by $\text{S}_\text{N}2$ mechanism :

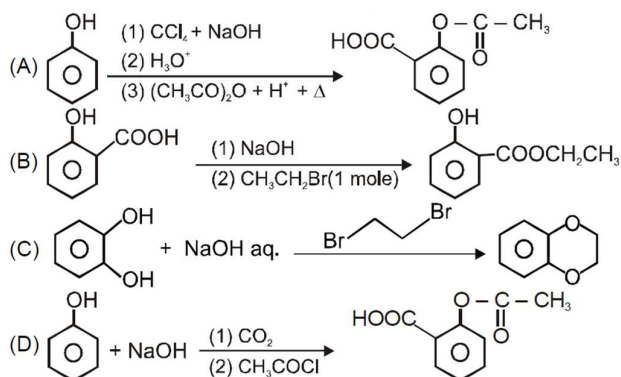
(A) $\text{CH}_3-\text{CH}_2-\text{ONa} + \text{CH}_3-\text{CH}_2-\text{Cl} \longrightarrow$

(B) + $\text{NaOH} \longrightarrow$

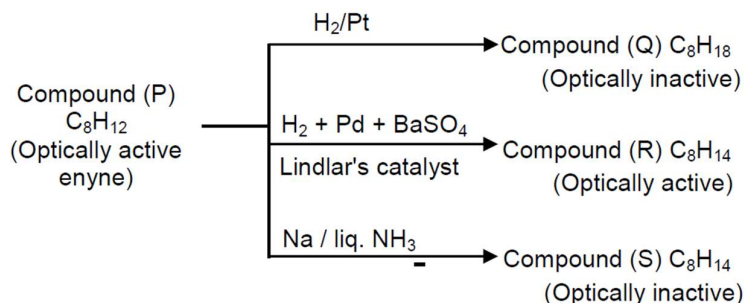
(C) $(\text{CH}_3)_3\text{C}-\text{CH}_2-\text{CH}_2-\text{Br} + \text{excess NH}_3 \longrightarrow$

(D) $\text{CH}_3-\text{CH}_2-\text{OH} + \text{HBr} \longrightarrow$

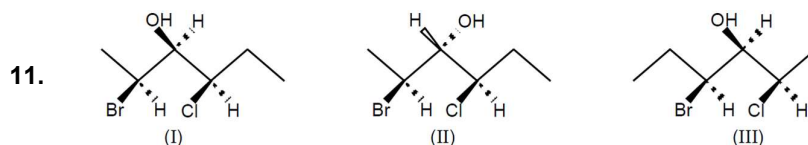
8. In which of the following reaction (s), reactant and products are correctly matched ?



9. The commonly used drug paracetamol is synthesized using the following sequence:
1. Nitration of phenol with dilute HNO_3 , producing mixture of X and Y. 6
 2. X is removed from the product mixture by steam-distillation.
 3. Reduction of Y by Sn/HCl to give Z.
 4. Acetylation of Z using 1 equivalent of acetic anhydride in presence of pyridine to give paracetamol.
- Which of the following is/are correct?
- (A) Paracetamol is antipyretic and analgesic (pain-killer).
 (B) X is ortho-nitrophenol
 (C) Z is ortho-amino phenol
 (D) During mono-acetylation of Z, $-\text{NH}_2$ group is acetylated in preference to $-\text{OH}$ group.
10. Which among the following is/are correct for above sequence of reactions ?



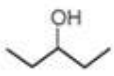
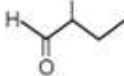
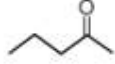
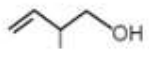
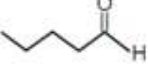
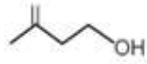
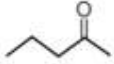
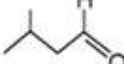
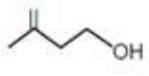
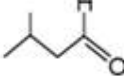
- (A) Oxidative ozonolysis product of P, R & S will be achiral
 (B) R & S are diastereomers
 (C) Product Q & R are formed by syn addition & S is formed by anti addition
 (D) P must have cis configuration



Choose the correct option/s according to given compounds.

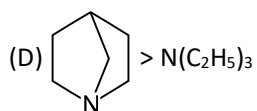
- (A) Compound (I)st & (III)rd are Identical
 (B) Compound (I)st & (II)nd are diastereomers
 (C) Compound (II)nd & (III)rd are positional Isomers
 (D) Compound (I)st & (III)rd are enantiomers
12. Three isomeric compound M, N and P ($\text{C}_5\text{H}_{10}\text{O}$) give the following tests:

- (i) M and P react with sodium bisulfite to form an adduct.
(ii) N consumes 1 mole of bromine and also gives turbidity with conc. HCl / anhydrous ZnCl₂ after prolong heating.
(iii) M reacts with excess of iodine in alkaline solution to give yellow crystalline compound with a characteristic smell.
(iv) p-Rosaniline treated with sulphur dioxide develops pink colour on shaking with P.
The structures of M, N and P, respectively can be:

	M	N	P
(A)			
(B)			
(C)			
(D)			

13. The correct order of nucleophilicity for following species is/are :

- (A) $\text{NH}_2^- > \text{NH}_3 > \text{Ph}-\ddot{\text{N}}\text{H}_2$
(B) $\text{Me}-\text{O}^- > \text{Me}_2\text{CH}-\text{O}^- > \text{Me}_3\text{C}-\text{O}^-$
(C) $\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$ (in polar protic solvent)



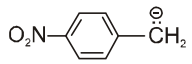
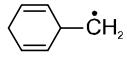
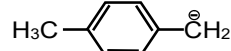
Match the column

Match the reactions given in List-I with the statements given in List-II.

LIST-I		LIST-II	
(I)	Ammonolysis	(P)	Amine with lesser number of carbon atoms
(II)	Gabriel phthalimide synthesis	(Q)	Detection test for primary amines.
(III)	Hoffmann Bromamide reaction	(R)	Reaction of phthalimide with KOH and R—X
(IV)	Carbylamine reaction	(S)	Reaction of alkylhalides with NH ₃

14. Which of the following has the correct combination considering List-I and List-II ?
(A) (I - S) (B) (II - R) (C) (III - P) (D) All of these
15. Which of the following combination reaction, carbene is used as electrophile ?
(A) (I - S) (B) (II - R) (C) (III - P) (D) (IV - Q)
16. Match the contents of List I with the contents of List II.

LIST-I

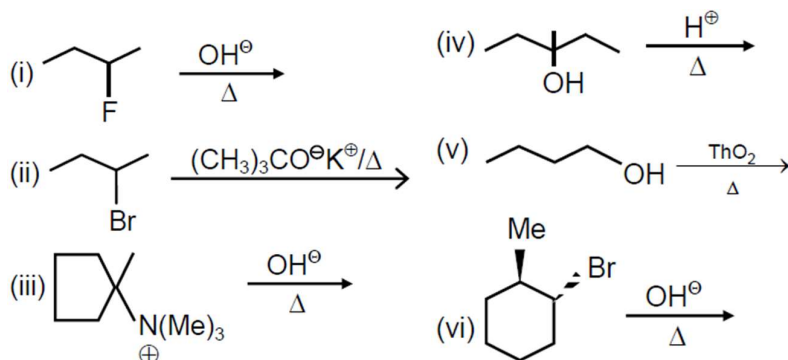
- (A) 
- (B) 
- (C) 
- (D) 

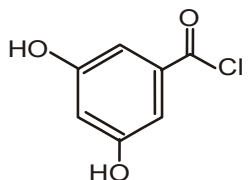
LIST-II

- (P) Stabilised by Hyperconjugation
- (Q) Destabilised by + I effect
- (R) Stabilised by + m effect
- (S) Stabilised by - m effect

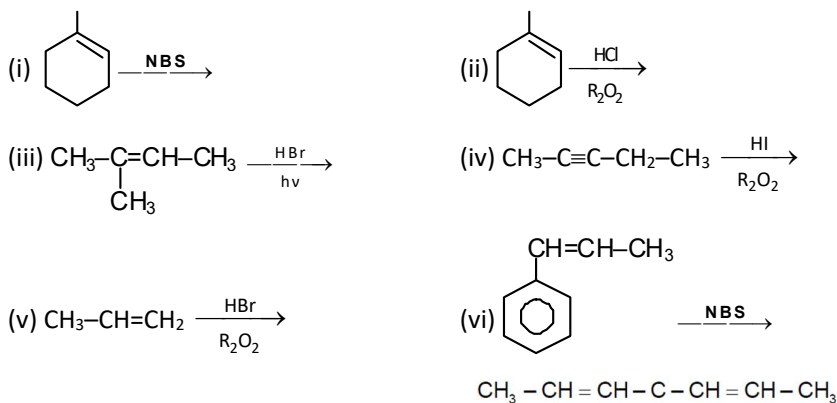
(Numerical Value type) :

17. Number of reactions in which Hoffman product (less stable alkene) formation takes place.



18.  Product is 3° Alcohol Find out value of 'x' :

19. How many reactions will proceed through free radical addition mechanism ?



20. How many stereoisomers are possible for

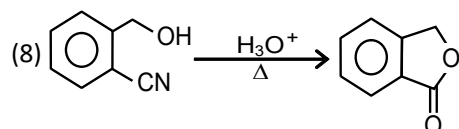
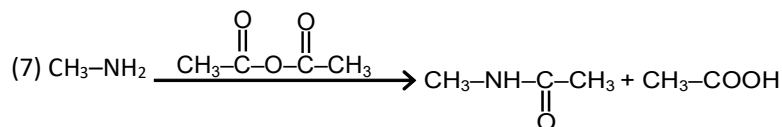
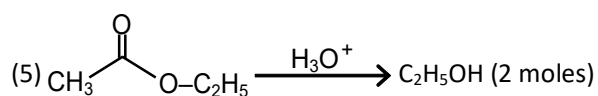
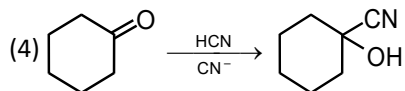
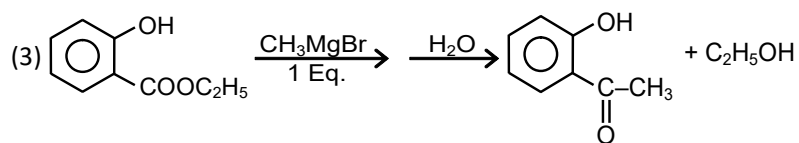
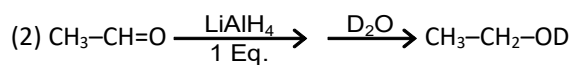
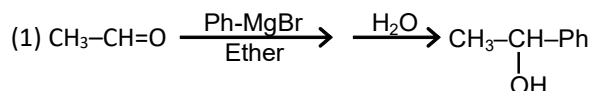


21. How many of the following are correctly matched :

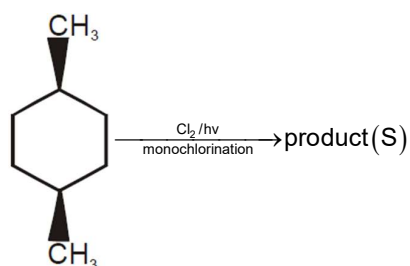
- S₁ : Nylon-6,6 $\longrightarrow$ Condensation polymer S₂ : Cellulose triacetate $\longrightarrow$ Natural polymer
- S₃ : Polyisoprene $\longrightarrow$ Homopolymer S₄ : Sucrose $\longrightarrow$ Polysaccharide
- S₅ : Starch $\longrightarrow$ Polymer of α -D-Glucose S₆ : Teflon $\longrightarrow$ Amide linkage

S7 : Nylon-6 $\longrightarrow$ Polymer of Caprolactum

22. In how many reactions correct major product is given :



23. How many optically active compounds are possible in the following reaction?



Answer Key

- | | | | | | | | | | |
|-----|---------------------------------------|-----|--------|-----|-------|-----|-------|-----|--------|
| 1. | (A) | 2. | (A) | 3. | (D) | 4. | (D) | 5. | (C, D) |
| 6. | (ABC) | 7. | (ABCD) | 8. | (ACD) | 9. | (ABD) | 10. | (ABC) |
| 11. | (BC) | 12. | (BD) | 13. | (ABD) | 14. | (D) | 15. | (D) |
| 16. | (A) – R , (B) – S , (C) – P , (D) – Q | | | | | 17. | 05.00 | 18. | 04 |
| 19. | 02.00 | 20. | 04.00 | 21. | 04.00 | | | | |
| 22. | 6 (1, 2, 4, 6, 7 & 8) | | | 23. | 4 | | | | |